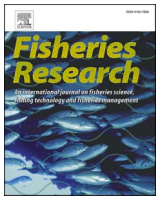
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An ensemble approach to understand predation mortality for groundfish in the Gulf of Alaska

Grant D. Adams [a,*], Kirstin K. Holsman [a,b], Steven J. Barbeaux [b], Martin W. Dorn [b], James N. Ianelli [b], Ingrid Spies [b], Ian J. Stewart [c], André E. Punt [a]

[a] School of Aquatic and Fishery Sciences, University of Washington, Seattle, WA, USA
[b] Resource Ecology and Fisheries Management Division, Alaska Fisheries Science Center, Seattle, WA, USA
[c] International Pacific Halibut Commission, Seattle, WA, USA

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ABSTRACT

There is increasing consensus of the need for ecosystem-based fisheries management (EBFM), which accounts for trophic interactions and environmental conditions when managing exploited marine resources. Continued development and testing of analytical tools that are expected to address EBFM needs are essential for guiding the management of fisheries resources in achieving and balancing multiple social, economic, and conservation objectives. To address these needs, we present and compare alternative climate-informed multi-species statistical catch-at-age models to account for spatio-temporal differences in stock distributions, with application to four groundfish species (walleye pollock *Gadus chalcogrammus,* Pacific cod *Gadus macrocephalus*, arrowtooth flounder *Atheresthes stomias*, and Pacific halibut *Hippoglossus stenolepis*) in the Gulf of Alaska, USA. We integrate across multiple forms of uncertainty regarding the data and distribution of Pacific halibut using an ensemble modelling approach. Models developed here can be used to supplement current tactical fisheries management and inform on the trade-offs between harvesting across groundfish in the Gulf of Alaska. This approach may be applicable for other situations where spatial and temporal overlap is extensive among closely coupled species.

1. Introduction

Changes in the population dynamics of exploited marine resources have long been understood to coincide with, or be the result, of trophic interactions (Hjort, 1914; Volterra, 1926). Yet, for various reasons, incorporation of ecosystem covariates and processes directly in stock assessments and management advice is not commonplace, although it is increasing (Dolan et al., 2016; Marshall et al., 2019; Skern-Mauritzen et al., 2015). Ecosystem Based Fisheries Management (EBFM) tools provide frameworks for quantifying ecological sand climate impacts on harvest recommendations, and is increasingly important in the context of climate change (e.g., Barbeaux et al., 2020). EBFM aims to manage fisheries resources by recognizing, in part, the trophic interactions among multiple fisheries (Dolan et al., 2016), for example, that fluctuations in the biomass of prey species can impact the abundance of predators (Pikitch et al., 2014) and vice-versa (Myers et al., 2007). EBFM approaches range from qualitative advice and inclusion of ecosystem information as context for near-term decision making to explicit incorporation of trophic and climate interactions into the management process (Fogarty and Rose, 2014; Howell et al., 2021; Levin et al., 2009; Marshall et al., 2018; Plagányi et al., 2022; Zador et al., 2017).

The continued development and testing of analytical tools that are expected to address EBFM needs is essential for guiding the management of fisheries resources in achieving and balancing multiple social, economic, and conservation objectives (Andersen et al., 2015; Deroba et al., 2015; Essington et al., 2018; Gaichas et al., 2017; Grüss et al., 2016; Punt et al., 2016). Multispecies and ecosystem models can directly account for trade-offs across objectives for multiple interdependent fisheries through explicit incorporation of trophic interactions (Fulton et al., 2019). Multispecies and ecosystem models range in complexity and application. At its most complex are well-known end-to-end ecosystem models such as Ecopath with Ecosim (Christensen and Walters, 2004) and Atlantis (Fulton et al., 2011) that model the energy pathways of entire food webs given parameter inputs. Such models are often used to support strategic, long-term management decision making by focusing

* Corresponding author.
E-mail address: adamsgd@uw.edu (G.D. Adams).

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on predicting broad scale changes across fisheries because of inherent structural uncertainty and a lack of fitting to observed data. At the less complex end of the range, are models of intermediate complexity for ecosystem assessments (MICE) (Plagányi et al., 2014), which estimate the ecological and population dynamics of a small number of interacting species to address specific management concerns. MICE strike a balance between single-species population dynamics models used to support tactical fisheries management decisions and end-to-end ecosystem models. MICE have particular utility in EBFM as they are statistically fit to data and provide estimates of parameter uncertainty, are designed to incorporate various ecological and ecosystem processes, and can be readily applied in an management strategy evaluation framework to quantify trade-offs among multiple species targeted by several fisheries (Plagányi et al., 2014).

Multispecies statistical catch-at-age analysis (MSCAA) is one type of MICE that has been developed to provide EBFM advice (Curti et al., 2013; Holsman et al., 2016; Jurado-Molina et al., 2005; Kinzey and Punt, 2009; Tsehaye et al., 2014; Van Kirk et al., 2010). Developed in part from the underpinnings of multispecies virtual population analysis (MSVPA; Jurado-Molina et al., 2005; Magnusson, 1995), MSCAA links single-species age-structured models through predation. The biological realism of these models comes at the cost of increased complexity, and possible increased uncertainty in estimated quantities (research on forecast performance is limited). Predation is conditioned on bio-energetics and diet information to parameterize consumption and the preference patterns of predators (Curti et al., 2013; Holsman et al., 2016; Kinzey and Punt, 2009). MSCAA can also estimate parameter uncertainty and account for process and observation error (Trijoulet et al., 2019). MSCAA can be parameterized similarly to single-species models used for tactical management and, therefore, are easily transitioned into current management frameworks. Similarly, MSCAA can also be used to derive fisheries reference points used for tactical management decision making (Gislason, 1999; Holsman et al., 2020; Moffitt et al., 2016).

MSCAA are particularly valuable for estimating time- and age-varying natural mortality and in some cases have been found to have improved predictive performance as compared to single species models that ignored predation (Trijoulet et al., 2019). However, MSCAA models may be particularly difficult to apply in situations where interacting populations do not share similar spatio-temporal distributions. For example, Pacific halibut *Hippoglossus stenolepis* are broadly distributed across the eastern Pacific Ocean and exhibit considerable interannual and seasonal movement (Carpi et al., 2021). However, food web analysis has indicated that Pacific halibut is highly connected to populations that inhabit much smaller ranges in the Gulf of Alaska (GOA), including: walleye pollock *Gadus chalcogrammus,* Pacific cod *Gadus macrocephalus*, and arrowtooth flounder *Atheresthes stomias* (Gaichas and Francis, 2008). The interannual and seasonal movement of Pacific halibut is hypothesized to have consequences for the dynamics of other groundfish species (Barnes et al., 2018). While there is interest in developing MSCAA in systems such as the GOA, misspecification of spatio-temporal distributions of populations can lead to bias in estimates of predation and biomass in MSCAA models (Kempf et al., 2010), limiting their utility for providing management advice. This highlights the need to develop models that account for variable spatio-temporal distributions among stocks to understand processes such as the historical and future impacts of Pacific halibut predation on groundfish stocks in the GOA.

Several approaches have been proposed to account for interdependent stocks that vary in their spatial overlap. Estimates of predator abundance have been multiplied by indices of predator-prey overlap to scale predation mortality (Kempf et al., 2010). This approach may be biased if data used to derive overlap indices are limited by the spatio-temporal distribution of sampling. For example, due to seasonality in the timing of the interannual migration of Pacific halibut through the GOA, actual overlap at shelf depths may differ between synoptic summer survey observations and cumulative annual patterns. Alternatively, predator biomass can be derived from survey data across the distribution of prey species and input into extended single-species models and MSCAA (e.g., Carroll et al., 2019; Spencer et al., 2016; van Kirk et al., 2012) or total predator abundance input into extended single-species models and scaled by an estimable proportionality constant (e.g., Hollowed et al., 2000). However, these approaches do not dynamically account for impacts of predator age- and size-structure on prey populations, or the influence of environmental conditions (e.g., temperature) on predation demand. While MSCAA are flexible enough to incorporate different approaches for accounting for variable spatio-temporal overlap of predator and prey, each approach will likely lead to differing perceptions of the dynamics of prey populations and should be evaluated relative to the data available.

As an alternative to developing a single "best" model, we developed a suite of models that differ in structure and/or data extent. We used this suite as an ensemble to provide a better representation of uncertainty in predation mortality arising from differences in spatio-temporal distributions of predator and prey populations (Millar et al., 2015; Hoeting et al., 1999; Stewart and Hicks, 2018). The ensemble consists of alternative MSCAA that account for spatio-temporal differences in stock distributions, with application to groundfish in the GOA. The core model was based on a climate-enhanced MSCAA model originally developed for groundfish in the Bering Sea (CEATTLE; Holsman et al., 2016). We adopted it to the GOA ecosystem and included key groundfish stocks (walleye pollock, Pacific cod, Pacific halibut, and arrowtooth flounder). CEATTLE (Climate-Enhanced, Age-based model with Temperature-specific Trophic Linkages and Energetics) uses fishery and survey data to estimate trends in biomass, predation, and recruitment. It is linked to climate via temperature-specific, bioenergetics-based predation functions. We expand on the original model to allow for multiple fisheries, surveys, sexes, selectivity functions, and the inclusion of fixed predator abundances (e.g., predators-as-fleets or "extended MSCAA") to account for the population dynamics of Pacific halibut. Additionally, sensitivity to how Pacific halibut predation is incorporated was explored by using an ensemble approach to average over, respectively, alternative parameterizations, data, and historical distributions.

2. Methods

2.1. Study area

The GOA is a topographically diverse ecosystem that supports many commercial fisheries whose target species exhibit strong trophic interactions (Zador et al., 2019). Walleye pollock, Pacific cod, Pacific halibut, and arrowtooth flounder are some of the most intensively sampled fish species in the GOA (Gaichas and Francis, 2008; Livingston et al., 2017)), and they support some of the largest fisheries in the region by volume (Fissel et al., 2020). For this study, we used the management areas for each species in the western and central GOA (Fig. 1). Pollock in the GOA is managed in National Marine Fisheries Service (NMFS) management areas 610, 620, 630, and 640 west of 140° W (Dorn et al., 2018). Pacific cod is managed in NMFS areas 610–659, with the majority (90%) of the biomass found west of 140° W (Barbeaux et al., 2018). Arrowtooth flounder is managed in NMFS areas 610–650, with the majority (85%) of the biomass found west of 140° W (Spies and Palsson, 2018). Pacific halibut is managed and assessed from California to the Bering Sea (Stewart and Hicks, 2019). However, given the majority of the biomass for the three other groundfish is found in the western and central GOA, we parametrized the model based on halibut from the IPHC Regulatory Areas 3 A and 3B (grey region, Fig. 1).

2.2. Assessment data

The MSCAA developed here is only fitted to data for walleye pollock, Pacific cod and arrowtooth flounder; the model is not fitted to data for Pacific halibut, the numbers-at-age over time for Pacific halibut are instead pre-specified. This approach was used to simplify estimation

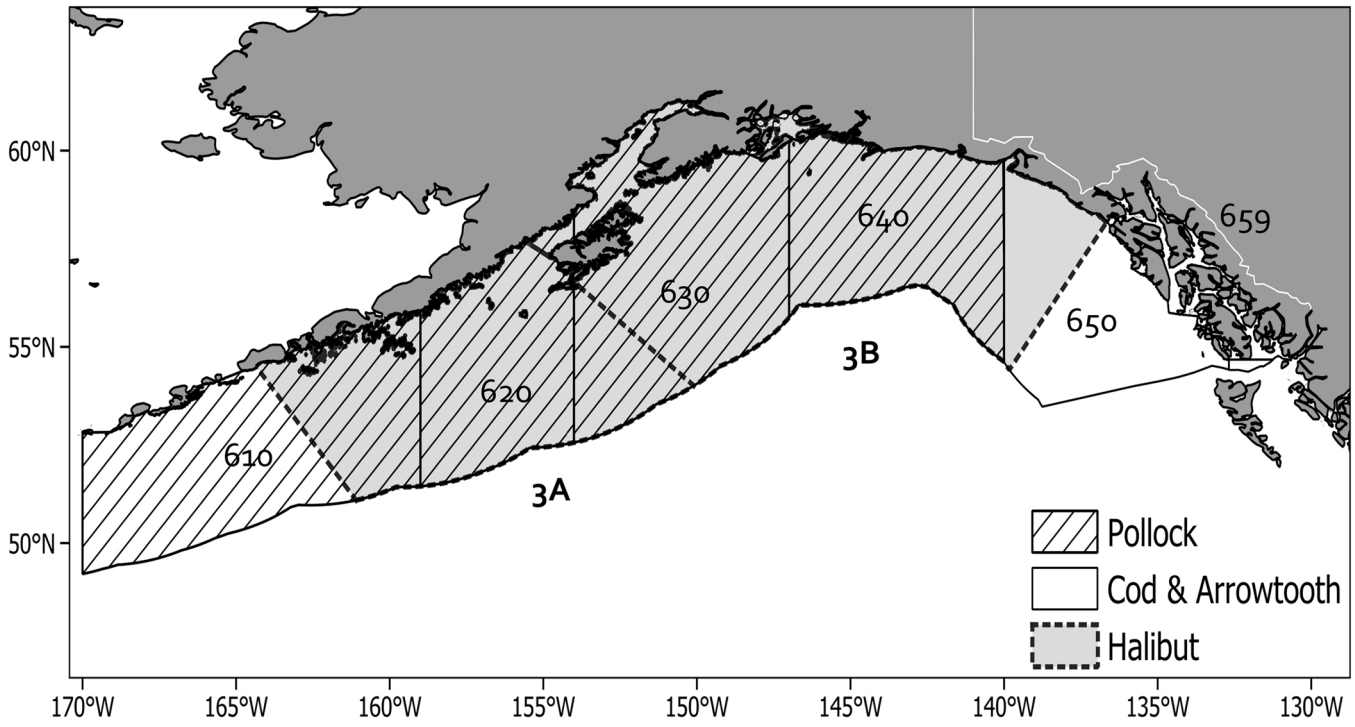


Fig. 1. Study area and stock boundaries of the GOA CEATTLE model. Included are IPHC Regulatory Areas 3A and 3B and NMFS reporting areas 610–659.

because Pacific Halibut are primarily predators in the system (Gaichas et al., 2011) and predation on halibut was not a focus of this study. Additionally, Pacific halibut dynamics are driven by environmental and density related factors over a much wider geographic range than the GOA (Clark and Hare, 2002). The data included in the likelihood function are indices of relative abundance or biomass and age- or length-composition data from the National Oceanic and Atmospheric Administration, Alaska Fishery Science Center (NOAA, AFSC) and Alaska Department of Fish and Game (ADFG), time series of total catch (metric tonnes) and fishery age- or length-composition data for these species. Index and composition data used for parameter estimation are the same as those used in recent stock-assessments (Barbeaux et al., 2018; Dorn et al., 2018; Spies and Palsson, 2018). Data input directly (i.e., not included in the likelihood function) include the mean weight-at-age, maturity-at-age, spawning month, ageing error, and age-length transition matrices, which were extracted from recent stock-assessments (Barbeaux et al., 2018; Dorn et al., 2018; Spies and Palsson, 2018).

Estimated mid-year numbers-at-age for Pacific Halibut from the most recent stock assessment and relative abundance-at-age from the IPHC set-line survey were used as model inputs. The assessment of Pacific halibut consists of multi-model averaging across four equally weighted models fit using Stock Synthesis (Methot and Wetzel, 2013) to estimate trends between 1996 and 2018 (Stewart and Hicks, 2019): long-time (1917–2018) and short-time (1996–2018) series models with either a single-fleet (coast-wide) or four-fleet configuration (areas-as-fleets). The areas-as-fleets approach is used to account for potential differences in selectivity and catchability of fisheries and surveys among regions, allowing the model to better account for spatio-temporal trends in data collection and fishing patterns. This avoids issues of extrapolating data to the coastwide level. Each of the models also differ in terms of the scale of natural mortality. Numbers-at-age from the four IPHC models represent the Pacific halibut stock from California to the Bering Sea. To scale the annual numbers-at-age to the GOA, the annual proportion of the halibut stock age 5 + by mass in IPHC Regulatory Areas 3A and 3B between 1993 and 2019 was taken from the IPHC fishery-independent setline survey (Stewart and Webster, 2022; Supplementary Fig. 1). The weight-at-age for halibut was also restricted to just IPHC Regulatory Areas 3A and 3B. Relative abundance-at-age estimates of halibut age 5 + from the IPHC fisheries independent set-line survey were used as a sensitivity analysis (see below). Very few Pacific halibut less than 5 years old are captured by the IPHC fisheries independent set-line survey and, therefore, therefore using age-5 + includes the broadest age range possible (Stewart and Webster, 2022).

2.3. Diet data

Data used to parameterize predation include temperature-specific predation ration from Holsman and Aydin (2015) and Holsman et al. (2019a), water temperature from the AFSC bottom trawl survey, and the average diet proportions by weight of predators-at-age for prey-at-age sampled from the AFSC bottom trawl survey (Livingston et al., 2017) (Table 1. The average diet proportion by weight of predators-at-age for prey-at-age ($\overline{U}_{pbjisa}$; Eq. 1 below) was estimated from stomach content data following an approach adapted from previous studies (Barnes et al., 2020; Stewart and Hamel, 2014; Supplementary material). $\overline{U}_{pbjisa}$ was assumed to pertain for all years to account for biennial sampling and a lack of standardized data prior to 1990 (Table 1; Equation 2.2) and averaged across data starting in 1990, 1993, or 1996–2015 depending on model variant (see below). While stomach content data were sampled prior to 1990, sampling was opportunistic and the diet data for the years prior to 1990 were consequently excluded (i.e., averages from stomach data for 1990–2015 were applied to these earlier years). The average diet proportion was estimated by averaging annual proportions of prey-at-age in the stomach of a predator-at-age across years relative to annual abundance-at-age:

$$\overline{U}_{pbjisa} = \frac{\sum_{y} \widehat{U}_{pbjisay} \widehat{I}_{py}}{\sum_{y} \widehat{I}_{py}} \quad (1)$$

where $\overline{U}_{pbjisa}$ is the average proportion of prey i of sex s and age a in the stomach of predator p of sex b and age j, $\widehat{U}_{pbjisay}$ is the annual proportion,

Table 1
Predation mortality equations for predators of species p, sex b, age j, and prey of species i, sex s and age a. Input ration parameters include: observed bottom temperature (T_y °C), Thermal optima for consumption (T_p^{co} °C), thermal limit of consumption (T_p^{cm} °C), rate at which consumption increased over relatively low water temperatures (Q_p^c), weight specific intercept (α_p^δ) and slope (β_p^δ) of maximum consumption, and scalar for maximum to observed consumption (φ_{pbjy}). For all parameter definitions see Table S1.

Definition	Equation	
Predation mortality	$M2_{isay} = \sum_{pbj}\left(\frac{\overline{N}_{pbjy}\delta_{pbjy}\widehat{S}_{pbjisa}}{\sum_{isa}\left(\widehat{S}_{pbjisa}\overline{B}_{isay}\right)+B_p^{other}\left(1-\sum_{isa}\left(\widehat{S}_{pbjisa}\right)\right)}\right)$	T2.1
Predator-prey suitability	$\widehat{S}_{pbjisa} = \frac{1}{n_y}\sum_{y}\left(\frac{\frac{\overline{U}_{psaisa}}{\overline{B}_{isay}}}{\sum_{isa}\left(\frac{\overline{U}_{pbjisa}}{\overline{B}_{isay}}\right)+\frac{1-\sum_{isa}\overline{U}_{pbjisa}}{B_p^{other}}}\right)$	T2.2
Individual specific ration ($kg\ kg^{-1}yr^{-1}$)	$\delta_{pbjy} = \varphi_{pbjy}\alpha_p^\delta W_{pbjy}^{\left(1+\beta_p^\delta\right)} f(T_y)_p$	T2.3
Temperature scaling algorithm	$f(T_y)_p = V^X e^{(X(1-V))}$	T2.4
…	$V = \left(T_p^{cm}-T_y\right)\Big/\left(T_p^{cm}-T_p^{co}\right)$	T2.4a
…	$X = \left(Z^2\left(1+(1+40/Y)^{0.5}\right)^2\right)\Big/400$	T2.4b
…	$Z = ln\left(Q_p^c\right)\left(T_p^{cm}-T_p^{co}\right)$	T2.4c
…	$Y = ln\left(Q_p^c\right)\left(T_p^{cm}-T_p^{co}+2\right)$	T2.4d

Source:Modified from Holsman et al. (2016) Table 2.

and $\widehat{I}_{py}$ relative abundance of predator p in year y estimated from the AFSC bottom trawl survey (see supplementary material for derivation). MSCAA models require data on the proportion of prey-at-age, by weight, in the stomach of predator-at-age relative to all other prey/ages while diet proportion data were obtained as weights categorized into predators-at-length and prey-at-length. Moreover, the AFSC uses a length-based stratified random sampling design for bottom trawl surveys in the GOA. $\widehat{U}_{pbjisay}$ was calculated from the annual length-based diet proportions (see supplementary material for calculation) by utilizing age-transition matrices to account for variability in length-at-age:

$$\widehat{U}_{pbjisay} = \frac{\sum_a\sum_k \widehat{U}_{pkily}ALK_{isaly}ALK_{pbjky}}{\sum_a\sum_k ALK_{isaly}ALK_{pbjky}} \quad (2)$$

where $\widehat{U}_{pkily}$ is the proportion of prey i of length l in the stomachs of predator p of length k in year y and ALK_{isaly} and ALK_{pbjky} are annual age-length transition matrices for prey species i and predator species p, respectively, used to account for interannual variability in length-at-age (see supplementary material for more information).

2.4. Population model

CEATTLE is an age-structured model that follows cohorts of individuals of a species, age, and sex through time (Table 2). Annual recruitment (Eq. T1.1.) and the initial age-structure (Eq. T1.2.) are estimated and are treated as random effects, with the variance in recruitment estimated. Walleye pollock and Pacific cod are treated as sex-combined while males and females are modelled separately for arrowtooth flounder and Pacific halibut to account for differential growth and sex-specific predation. The sex-ratio at birth is assumed to be 0.5 for arrowtooth flounder and Pacific halibut. Natural mortality (M) can be input, estimated, or partitioned into predation mortality (M2) and non-predation related mortality (M1) (Eq T1.11 and T1.13), the latter of which is either input or estimated, and accounts for mortality related to causes other than predation by the species included in the model.

2.5. Predation (M2)

Predation mortality is modelled following previous MSCAA models (Curti et al., 2013; Holsman et al., 2016; Tsehaye et al., 2014) based on MSPVA. Predation mortality due to a predator-at-age is derived from empirically derived suitability coefficients, estimates of annual consumption, estimated or input biomass, and input biomass of other prey in the system (B_p^{other})(Table 1). Suitability represents a predator's prey and size preference and factors such as spatial overlap in the distribution of predators and prey. Suitability is derived empirically from the ratio of mean proportion (by weight) of prey-at-age in the stomach of predators-at-age (Eq.1; Table 1). Magnusson (1995) provides details on the derivation of suitability coefficients. Bioenergetics-based, temperature- and age-dependent consumption rates (g/g/d) were based on previous studies (Holsman et al., 2019a; Holsman and Aydin, 2015). Estimated average biomass is used to derive predation mortality and because of the interdependence between predation and biomass, predation mortality and the underlying age-structured dynamics was iteratively estimated until convergence occurred. Preliminary analysis suggested that three iterations were sufficient for convergence. Models were insensitive to a range of $B_p^{other}$and therefore, we, utilized values from Holsman et al. (2016), which were derived from ecosystem-wide biomass estimates for the Eastern Bering Sea.

2.6. Observation model (walleye pollock, Pacific cod, arrowtooth flounder)

CEATTLE was fit to time series of fishery and survey biomass, sex-ratio, and length- or age-composition data. Log-fishery catch, log-survey biomass and sex-ratio were assumed to be normally distributed, while age- and length-composition data were fit assuming multinomial distributions (Table 3). Variance parameters for the lognormal distributions and initial input sample size for the multinomial distributions were assumed known. Separate selectivity and catchability functions were estimated for each survey or fishery (Table 4). For walleye pollock and arrowtooth flounder CEATTLE was parameterized almost exactly as in recent stock assessments (Dorn et al., 2018; Spies and Palsson, 2018). The implementation of CEATTLE for walleye pollock differed from the actual assessment in that lognormal recruitment deviates for all years rather than only the foremost and terminal years were penalized assuming a normal distribution and known or estimated variance. The implementation for arrowtooth flounder differed from the actual assessment in that non-parametric selectivity penalties for males were assumed to be the same as for females. The Pacific cod assessment was simplified to use marginal length-composition data rather than conditional age-at-length, and selectivity functions were selected to mimic the general form of selectivity in the Pacific cod assessment, while limiting the use of double normal time-varying selectivity (Table 4). Comparison of time-series of biomass and spawning stock biomass between single-species CEATTLE models and the actual stock assessments showed agreement when CEATTLE ignored predation (Supplementary Fig. 2).

2.7. Including halibut in the model

Rather than estimate the numbers-at-age for both sexes of Pacific halibut as part of the model fitting process, estimated numbers-at-age for both sexes were directly input into the model (Eq. T1.3.). Numbers were either provided as absolute or relative numbers-at-age. In the former case, numbers-at-age were calculated as the numbers-at-age for both sexes at the mid-point of the year from the four IPHC sub-models multiplied by the annual proportion of the halibut stock age 5 + in

Table 2
General population dynamics equations for species i, sex s, age a, and length l in year y, and the observation and fishery model for survey or fishery f_i. For parameter definitions see Table S1.

Definition	Equation	
Recruitment	$N_{is1y} = R_{isy} = \overline{R}_i e^{\tau_{iy}} \rho_{i1}$	T1.1
Initial abundance	$N_{isa1} = \begin{cases} \overline{R}_i e^{\left(-\sum_{j=1}^{a-1} M1_{isj}+\tau_{ia}\right)} \rho_{i1} & \text{if} j < A_i \\ \dfrac{\overline{R}_i e^{\left(-\sum_{j=1}^{a-1} M1_{isj}+\tau_{ia}\right)} \rho_{i1}}{\left(1-e^{\left(-M1_{isA_i}\right)}\right)} & \text{if} j = A_i \end{cases}$	T1.2
Numbers at age	$\begin{cases} \dfrac{N_{isa+1y+1} = N_{isay} e^{-Z_{isay}}}{N_{isA_i y+1} = N_{isA_i-1y} e^{-Z_{isA_i-1y}} + N_{isA_i y} e^{-Z_{isA_i y}}} \text{if i} \neq \text{Halibut} \\ N_{isay} = d_{iy} * N_{Input.isay} \quad \text{if i = Halibut and model} \neq 11 \\ N_{isay} = \widetilde{q}_i * \widehat{I}_{isay} \quad \text{if i = Halbiut and model = 11} \end{cases}$	T1.3
Average numbers at age	$\overline{N}_{isay} = \dfrac{N_{isay} * \left(1-e^{-Z_{isay}}\right)}{Z_{isay}}$	T1.4
Survey biomass (kg)	$\widehat{B}_{f,y} = \sum_s \sum_a N_{isay} e^{-Month_{f_i} * Z_{isay}} W_{f,say} sel_{f,say} q_{f,y}$	T1.5
Survey abundance (numbers)	$\widehat{I}_{f,y} = \sum_s \sum_a N_{isay} e^{-Month_{f_i} * Z_{isay}} sel_{f,say} q_{f,y}$	T1.6
Catch-at-age	$C_{fisay} = \dfrac{F_{fisay}}{Z_{isay}} \left(1-e^{-Z_{isay}}\right) N_{isay}$	T1.7
Total catch (kg)	$\widehat{C}^*_{f,y} = \sum_s \sum_j \dfrac{F_{fisay}}{Z_{isay}} \left(1-e^{-Z_{isay}}\right) N_{isay} W_{fisay}$	T1.8
Biomass at age (kg)	$B_{isay} = N_{isay} W_{Bia y}$	T1.9
Spawning biomass at age (kg)	$SB_{iay} = \begin{cases} N_{iay} e^{-Month_{ssb_i}/12 * Z_{iay}} W_{ssb_i ay} \rho_{ia} \delta_{ia} & \text{if1sex model} \\ N_{i1ay} e^{-Month_{ssb_i}/12 * Z_{i1ay}} W_{ssb_i 1ay} \delta_{ia} & \text{if2sex model} \end{cases}$	T1.10
Total mortality at age	$Z_{isay} = M1_{isa} + M2_{isay} + \sum_{f_i} F_{fisay}$	T1.11
Fishing mortality at age and fleet	$F_{fisay} = \overline{F}_{f_i} e^{\epsilon_{f,y}} sel_{fisay}$	T1.12
Residual/natural mortality	$M1_{isj} = \begin{cases} M1_{isj} & \text{if single-species} \\ \widehat{M1}_{is} & \text{if mult-species} \end{cases}$	T1.13
Predicted age composition	$\widehat{O}_{fi\widehat{say}} = \dfrac{\sum_a C_{fisay} AE_{i\widehat{aa}}}{\sum_s \sum_{\widehat{a}} \sum_a C_{fisay} AE_{i\widehat{aa}}} \text{if} f_i = \text{fishery}$	T1.14
	$\widehat{O}_{fi\widehat{say}} = \dfrac{\sum_a N_{isay} e^{-Month_{f_i} * Z_{isjy}} sel_{fisay} AE_{i\widehat{aa}}}{\sum_s \sum_{\widehat{a}} \sum_a N_{isay} e^{-Month_{f_i} * Z_{isjy}} sel_{fisay} AE_{i\widehat{aa}}} \text{if} f_i = \text{survey}$	
Predicted length composition	$\widehat{O}_{fisly} = \dfrac{\sum_{\widehat{a}} \sum_a C_{fisay} AE_{i\widehat{aa}} ATM_{fisal}}{\sum_s \sum_l \sum_{\widehat{a}} \sum_a C_{fisay} AE_{i\widehat{aa}} ATM_{fisal}} \text{if} f_i = \text{fishery}$	T1.15
	$\widehat{O}_{fisly} = \dfrac{\sum_{\widehat{a}} \sum_a N_{isay} e^{-Month_{f_i} * Z_{isay}} sel_{fisay} AE_{i\widehat{aa}} ATM_{fisal}}{\sum_s \sum_l \sum_{\widehat{a}} \sum_a N_{isay} e^{-Month_{f_i} * Z_{isjy}} sel_{fisay} AE_{i\widehat{aa}} ATM_{fisal}} \text{if} f_i = \text{survey}$	

management area 3 (d_{iy}) from IPHC fisheries independent set-line survey (Stewart and Webster, 2022). In the case of relative numbers-at-age, relative abundance-at-age for both sexes of halibut age 5 + in IPHC Regulatory Areas 3A and 3B from the IPHC fisheries independent set-line survey ($\widehat{I}_{isay}$; Stewart and Webster, 2022) were multiplied by an estimable parameter ($\widetilde{q}_i$) to scale the relative-numbers-at-age to numbers-at-age that may be relevant for predation. We assumed a linear relationship between relative-abundance-at-age and abundance-at-age following the Pacific halibut stock assessment. No data for Pacific halibut are included in the likelihood function so any estimate of $\widetilde{q}_i$ is based on the fit to the data for the remaining species.

2.8. Model variants

Sixteen model variants were constructed to evaluate the impact of alternative ways to include predation by Pacific halibut in the assessment. Specifically, models were fit that excluded and included predation mortality and evaluated alternative assumptions regarding the distribution and abundance of Pacific halibut. To account for time differences in data availability (Table 5), eight long time-series models (models 1–8) were constructed that spanned 1977–2018; three medium time-series models (models 9–11) were constructed that spanned 1993–2018; and five short time-series models (models 12–16) were constructed that spanned 1996–2018. The long time-series models represent the longest time-series for which data were available for all species and utilizes diet proportion data averaged over 1990 and 2015 (Eq. 1). The medium time series models account for the period when relative abundance-at-age data are available for Pacific halibut and utilize diet proportion data averaged over 1993 and 2015. The short time-series models account for the period during which mid-year numbers-at-age of Pacific halibut were available from the short-term models in the multimodel assessment (estimates of mid-year numbers-at-age were unavailable for the start year of the model) and we averaged diet proportion data from 1996 through 2015. Models were constructed using estimates of the mid-year numbers-at-age from each of the four Pacific halibut models (long- and short time-series coastwide and areas-as-fleets models) to account for differences in the assumptions and data used in the Pacific halibut assessment (Stewart and Hicks, 2019). As mentioned above, numbers-at-age of Pacific halibut were scaled to the GOA using the relative proportion of age-5 + halibut (based on selectivity of the survey) in IPHC Regulatory Areas 3A and 3B between 1993 and 2018. Alternative assumptions regarding the distribution of halibut in the GOA prior to 1993 are described below and in Table 5.

The long time-series models for 1977–2018 were:

Table 3
Components of the likelihood function for species *i*, sex *s*, and age *a* during year *y* for survey *s* or fishery *f*. For parameter definitions see Table S1.

Description	Equation	
Data components		
Survey biomass	$ln(B_{f,y}) \sim \mathrm{N}\left(ln(\widehat{B}_{f,y}) - \frac{\sigma^2_{f,y}}{2}, \sigma^2_{f,y}\right)$	T3.1
Total catch	$ln\left(C^*_{f,y}\right) \sim \mathrm{N}\left(ln\left(\widehat{C}^*_{f,y}\right) - \frac{\sigma^2_{f,y}}{2}, \sigma^2_{f,y}\right)$	T3.2
Age composition	$-\sum_i \sum_s n_{f,s,y} \sum_a \sum_y \left(\widehat{O}_{f,s,\widehat{a},y} + 0.00001\right) * ln\left(O_{f,s,\widehat{a},y} + 0.00001\right)$	T3.3
Length composition	$-\sum_i \sum_s n_{f,s,y} \sum_a \sum_y \left(\widehat{O}_{f,s,l,y} + 0.00001\right) * ln\left(O_{f,s,l,y} + 0.00001\right)$	T3.4
Sex ratio	$\frac{\sum_a N_{i1,a,y}}{\sum_s \sum_a N_{i,s,a,y}} \sim \mathrm{N}\left(\overline{\rho}_i, \sigma^2_{\rho_i}\right)$	T3.5
Penalties		
Non-parametric selectivity	$\sum_i \sum_s \sum_a^{A_i - 1} \chi_{f_i} \left[ln\left(\frac{e^{\psi_{f_i,s,a}}}{e^{\psi_{f_i,s,a+1}}}\right) - ln\left(\frac{e^{\psi_{f_i,s,a+1}}}{e^{\psi_{f_i,s,a+2}}}\right)\right]^2 \quad \chi_{f_i} =$ 0if $e^{\psi_{f_i,s,a}} \leq e^{\psi_{f_i,s,a+1}}$	T3.6
Recruitment deviate	$\tau_{i,y} \sim \mathrm{N}\left(-\frac{\sigma^2_{R_i}}{2}, \sigma^2_{R_i}\right)$	T3.7
Fishing mortality deviate	$\epsilon_{f,i,y} \sim \mathrm{N}\left(-\frac{\sigma^2_{f_i}}{2}, \sigma^2_{f_i}\right)$	T3.8
Annual selectivity deviate	$\psi_{f,i,y} \sim \mathrm{N}\left(-\frac{\sigma^2_{\psi_{f_i}}}{2}, \sigma^2_{\psi_{f_i}}\right)$	T3.9
Mean catchability priors	$\overline{q}_{f_i} \sim \mathrm{N}\left(\widetilde{q}_{f_i} - \frac{\sigma^2_{q_{f_i}}}{2}, \sigma^2_{q_{f_i}}\right)$	T3.10
Catchability deviates 1 (random deviates)	$\omega_{f,i,y} \sim \mathrm{N}\left(-\frac{\sigma^2_{\omega_{f_i}}}{2}, \sigma^2_{\omega_{f_i}}\right)$	T3.11a
Catchability deviates 2 (random walk)	$\omega_{f,i,y+1} \sim \mathrm{N}\left(\omega_{f,i,y} - \frac{\sigma^2_{\omega_{f_i}}}{2}, \sigma^2_{\omega_{f_i}}\right)$	T3.11b

Source:Modified from Holsman et al. (2016).

- Model 1: a model that did not include predation (i.e., a set of single-species models), representing a base model.
- Model 2: a model that did not include halibut predation to allow comparisons in which halibut does not impact the dynamics of groundfish in the GOA.
- Models 3–5: models that included pre-specified mid-year numbers-at-age of Pacific halibut from the coastwide long-time (1917–2018) series model developed by the IPHC. To account for a lack of information on halibut distribution prior to 1993, numbers-at-age prior to 1993 were multiplied by the 50th (model 3), 15th (model 4), and 85th (model 5) quantiles of the proportion of adult halibut in area 3 between 1993 and 2018 assuming no temporal variation prior to 1993 (Supplementary Fig. 1).
- Models 6–8: as for models 3–5 but using numbers-at-age of Pacific halibut from the areas-as-fleets long-time series model.

The three medium term models for 1993–2018 were:

- Model 9: a set of single-species models to represent a base single-species model.
- Model 10: a model that did not include halibut predation.
- Model 11: a model with relative abundance-at-age of Pacific halibut in area 3 multiplied by an estimated parameter to allow the model to estimate the relative contribution of Pacific halibut predation to describe the dynamics of pollock, Pacific cod, and arrowtooth flounder.

Table 4
Overview of the parameterization of the survey and fishery fleets and the data sources used for GOA CEATTLE. For more detailed information on data refer to stock assessment reports (Barbeaux et al., 2018; Dorn et al., 2018; Spies and Palsson, 2018; Stewart and Hicks, 2019). Parenthesis refer to specific equations in Table S2. Note: all surveys except the Shelikof Strait winter acoustic survey are conducted during the summer.

Fleet	Data type	Selectivity	Catchability
Pollock:			
Shelikof Strait acoustic survey	Age-1 relative abundance	Fixed: $sel_{f,i,1y} = 1$	Time-invariant (ST2.9)
	Age-2 relative abundance	Fixed: $sel_{f,i,2y} = 1$	Time-invariant (ST2.9)
	Age-3 + relative biomass	Descending logistic (ST2.7)	Time-varying random walk (ST2.10; T3.11b)
	Age composition		
NMFS bottom trawl survey	Relative biomass	Logistic (ST2.2)	Time-invariant (ST2.9)
	Age composition		Mean N(0.85, 0.1) prior (T3.10)
ADFG trawl survey	Relative biomass	Logistic (ST2.2)	Time-varying random walk (ST2.10; T3.11b)
	Age composition		
Summer acoustic survey	Relative biomass	Fixed: $sel_{f,i,a,y} = 1$	Time-invariant (ST2.9)
	Age composition		
Pollock fishery	Total catch	Time-varying double logistic 2 (ST2.6)	
	Age composition		
Arrowtooth flounder:			
NMFS bottom trawl survey	Relative biomass	Logistic (ST2.2)	Fixed $q_{f_i} = 1$
	Length composition		
	Age composition		
Fishery	Total catch	Non-parametric (ST2.21)	
	Length composition		
Pacific cod:			
NMFS bottom trawl survey	Relative abundance	Logistic (ST2.2)	Time-invariant (ST2.9)
	Age composition		
	Length composition		
NMFS sablefish longline survey	Relative abundance	Non-parametric (ST2.1)	Time-invariant (ST2.9)
	Length composition		
Trawl fishery	Total catch	Logistic (ST2.2)	
	Length composition		
Longline fishery	Total catch	Logistic (ST2.2)	
	Length composition		
Pot fishery	Total catch	Non-parametric (ST2.1)	
	Length composition		
Pacific halibut:			
IPHC longline survey	Relative abundance-at-age		

Table 5
Overview of the GOA CEATTLE model variants and ensemble model weights. Mode refers to whether estimation included predation mortality (single- vs multi-species) and halibut model refers to the assessment model from the multi-model ensemble used for the input numbers-at-age of Pacific halibut. Historical distribution refers to the quantiles of percentage of pre-1993 adult halibut in the GOA relative to 1993–2018.

Model	Model Avg (weight)	Start year	Mode	Halibut model	Historical distribution	JNLL†	K‡	AIC	ΔAIC	Wi
1	1 (0.33)	1977	Single-	None		68,378	605	137,967	1	0.35
2	**2 (0.33)**	**1977**	**Multi-**	**None**		**68,374**	**609**	**137,965**	**0**	**0.65**
3	3 (0.83)	1977	Multi-	Coastwide long	50%	68,384	609	137,987	21	0.00
4	3 (0.83)	1977	Multi-	Coastwide long	15%	68,384	609	137,987	21	0.00
5	3 (0.83)	1977	Multi-	Coastwide long	95%	68,384	609	137,987	22	0.00
6	3 (0.83)	1977	Multi-	AAF* long	50%	68,384	609	137,985	20	0.00
7	3 (0.83)	1977	Multi-	AAF* long	15%	68,384	609	137,985	20	0.00
8	3 (0.83)	1977	Multi-	AAF* long	95%	68,384	609	137,986	20	0.00
9	1 (0.33)	1993	Single-	None		53,168	447	107,230	14	0.00
10	**2 (0.33)**	**1993**	**Multi-**	**None**		**53,157**	**451**	**107,216**	**0**	**0.73**
11	4 (0.5)	1993	Multi-	Relative abundance		53,157	452	107,218	2	0.27
12	1 (0.33)	1996	Single-	None		45,817	411	92,455	0	0.40
13	**2 (0.33)**	**1996**	**Multi-**	**None**		**45,813**	**415**	**92,455**	**0**	**0.41**
14	3 (0.25)	1996	Multi-	Coastwide short		45,816	415	92,461	6	0.02
15	3 (0.25)	1996	Multi-	AAF* short		45,816	415	92,463	8	0.01
16	4 (0.5)	1996	Multi-	Relative abundance		45,813	416	92,457	2	0.16

* AAF = areas-as-fleets; † JNLL = joint negative log-likelihood, ‡ K = Number of fixed and random effects parameters.

The five short term models for 1996–2018 were:

- Model 12: a set of single-species models to represent a base single-species model.
- Model 13: a model that did not include halibut predation.
- Model 14: a model with pre-specified mid-year numbers-at-age of Pacific halibut from the coastwide short-time series model.
- Model 15: as for model 14, but using numbers-at-age for Pacific halibut from the areas-as-fleets short-time series model
- Model 16: as for model 11, but using relative abundance-at-age of Pacific halibut multiplied by an estimated scaling parameter, but with the model starting in 1996.

2.9. Model comparison and model averaging

Individual models were fit using Template Model Builder (TMB; Kristensen et al., 2015) and were compared using the Akaike Information Criterion (AIC; Burnham and Anderson, 2004) and associated delta AIC (ΔAIC) and AIC weights (Wi) and by evaluating the fit to each data source. In age-structured stock assessments, AIC is sensitive to the relative contribution of each likelihood component and tuning weights can be used to reduce conflicts between different data sources (Francis, 2011). Therefore, we used the tuned sample sizes for arrowtooth flounder and walleye pollock and tuned the sample size of marginal age- and length- composition data of Pacific cod using the harmonic mean of the effective sample size from the McAllister and Ianelli method (McAllister and Ianelli, 1997; Punt, 2017). Composition weights were tuned using the single-species models and held at those values for all multispecies models. AIC was compared separately for the long (1977–2018), medium (1993–2018), and short time series models (1996–2018) given the differences in likelihood between those sets of models. Similarly, multispecies models are sensitive to the value of residual natural mortality (M1), so we estimated age-invariant M1 for each species and sex across all models that included predation. For models that excluded predation, age-variant total natural mortality (M) was fixed at the value input or estimated (in the case of Pacific cod) in recent assessments (Supplementary Table 3; Barbeaux et al., 2018; Dorn et al., 2018; Spies and Palsson, 2018).

We used an ensemble modelling approach to account for uncertainty arising from the time-series of data used, the distribution of Pacific halibut in the GOA, and in the numbers-at-age of Pacific halibut (arising from treatments of historical data and spatial dimension of the data used for the ensemble halibut model; Stewart and Hicks, 2019). Four ensemble models were developed assuming equal weights for numbers-at-age of Pacific halibut used in the individual model: Avg 1) a single-species ensemble model (Models 1, 9, and 12), Avg 2) a multi-species ensemble model assuming no halibut predation (Models 2, 10, and 13), Avg 3) a multi-species ensemble model including halibut predation derived in part from pre-specified mid-year numbers-at-age of each of the four halibut sub-models (Models 3–8 and 14–15), and Avg 4) a multi-species ensemble model including halibut predation derived in part from relative numbers-at-age (Models 11 and 16). For Avg 3, models 3–5 and 6–8 were collectively given the same weight as models 14–15. Confidence intervals for derived quantities from ensemble models were estimated via a simulation approach. Parameter sets were simulated from a multivariate normal distribution with a mean of the maximum likelihood estimated parameters and variance of the inverse of the estimated joint Hessian matrix. Estimates of derived quantities of interest (recruitment, biomass, spawning stock biomass) were then calculated for each parameter set. Confidence intervals were then calculated across sets of derived quantities across sub-models.

3. Results

3.1. Fit to data

Three single-species and 13 multi-species age-structured models were fit to fishery and survey data for walleye pollock, arrowtooth flounder, Pacific cod, and Pacific halibut data in the GOA. Models that included predation mortality, but excluded predation from Pacific halibut led to the lowest AIC for all time-series (Tables 5–6). For the long and medium time-series models (Models 1–8), single-species models provided the best fit overall to the composition data (Table 6). Alternatively, for the long and medium time-series models (Models 9–11), models that included predation provided the best fit to the index data (Table 6). For short-time series models (Models 12–16), the single-species model (Model 12) led to the best fit to the index data (model 12), while the model that included predation (Model 13) led to the best fit to the composition data and the two models had a comparable AIC.

3.2. Comparison of alternative time series length from individual models

The long-, medium-, and short-term multi-species models fit with different Pacific halibut numbers-at-age had similar trends in population quantities for some time periods (Supplementary Figs. 3–5). However, the long-term models differed slightly between short- and medium-term models. For example, recent estimates of biomass and spawning stock biomass of walleye pollock and arrowtooth flounder were estimated to

Table 6
Overview GOA CEATTLE model variants and negative log-likelihood components. Mode refers to whether estimation included predation mortality (single- vs multi-species) and halibut model refers to the assessment model from the multi-model ensemble used for the input numbers-at-age of Pacific halibut. Historical distribution refers to the quantiles of percentage of pre-1993 adult halibut in the GOA relative to 1993–2018.

Model	Start year	Mode	Halibut model	Historical distribution	Index data	Total catch	Composition data	Rec devs	Sel pen	q pen	F dev
1	1977	Single-	None		158	-411	68,259	229	113	-149	167
2	1977	Multi-	None		143	-410	68,268	227	109	-147	156
3	1977	Multi-	Coastwide long	50%	143	-410	68,271	230	110	-147	157
4	1977	Multi-	Coastwide long	15%	143	-410	68,271	230	110	-147	157
5	1977	Multi-	Coastwide long	95%	143	-410	68,271	230	110	-147	157
6	1977	Multi-	AAF* long	50%	143	-410	68,271	230	110	-147	157
7	1977	Multi-	AAF* long	15%	143	-410	68,271	230	110	-147	157
8	1977	Multi-	AAF* long	95%	143	-410	68,271	230	110	-147	157
9	1993	Single-	None		144	-266	53,053	89	137	-83	48
10	1993	Multi-	None		143	-266	53,054	84	132	-82	45
11	1993	Multi-	Relative abundance		143	-266	53,054	84	132	-82	45
12	1996	Single-	None		139	-234	45,705	77	115	-72	36
13	1996	Multi-	None		147	-234	45,697	75	114	-70	37
14	1996	Multi-	Coastwide short		147	-234	45,697	75	115	-70	37
15	1996	Multi-	AAF* short		147	-234	45,698	75	115	-70	37
16	1996	Multi-	Relative abundance		147	-234	45,697	75	114	-70	37

* AAF = areas-as-fleets

be slightly greater in 1996 from long time-term multi-species models (Supplementary Figs. 3–4). While we estimated that arrowtooth flounder consumed the most biomass of walleye pollock in all multi-species models, the short-term models (Models 13–16) estimated that arrowtooth flounder consumed more biomass of walleye pollock than the long- and medium-term models (Supplementary Fig. 7). Cannibalism of arrowtooth flounder also differed among the long-, medium-, and short-term models (Supplementary Fig. 7) and total natural mortality (M1 + M2) was estimated to be lower in the long- versus short-term models for walleye pollock (Supplementary Fig. 9).

3.3. Comparison of alternative halibut distributions from individual models

For the long time-series models, estimates of biomass, spawning stock biomass, and recruitment varied little among assumptions regarding historical Pacific halibut distribution in the GOA pre-1993 (Supplementary Figs. 3–5). 15th, 50th, or 85th quantile of adult halibut in area 3 between 1993 and 2018 was assumed for pre-1993 abundance in area 3 led to little difference in derived quantities. Similarly, using the numbers-at-age from the Pacific halibut areas-as-fleets and coast-wide models provided very similar trends in derived

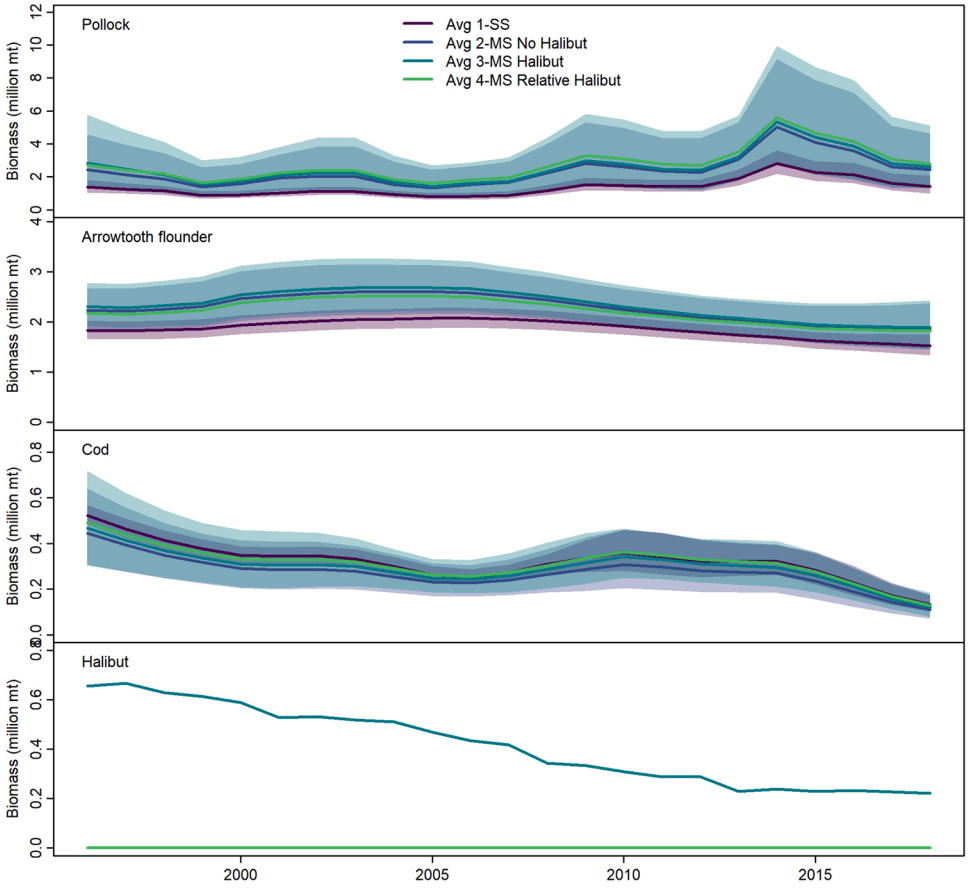


Fig. 2. Estimates of age-1+ total biomass (solid line) and 95% confidence interval (shaded region) from ensemble GOA CEATTLE Models. See Table 5 for model descriptions. Note: total biomass for Pacific halibut from Avg 4 is calculated from relative abundance-at-age for both sexes of halibut age 5+ in IPHC Regulatory Areas 3A and 3B from the IPHC fisheries independent set-line survey multiplied by an estimable scaling parameter (See Eq. T1.3, T1.9, and T1.10). Confidence intervals for Avg 4 are also excluded given unrealistically high uncertainty.

quantities for all individual models.

3.4. Time series of derived quantities from ensemble models

Ensemble models were constructed by averaging individual model results across predation types ignoring AIC (no predation, no halibut predation, halibut predation vis absolute or relative numbers at age; Table 5). There were relatively large differences in estimates of biomass, spawning stock biomass, and recruitment of walleye pollock and arrowtooth flounder relative to the single-species ensemble model for ensemble models that included predation and estimated residual natural mortality (Figs. 2–3). For example, across all years, ensemble models that included predation estimated biomass of pollock to be on average 2.03 (Avg 2), 2.21 (Avg 3), and 2.07 (Avg 4) times larger than estimates from the single-species ensemble model. Similarly, spawning stock biomass of walleye pollock was estimated to be 1.70 (Avg 2), 1.84 (Avg 3), and 1.71 (Avg 4) times greater across all years when predation was included. These differences were also evident when comparing individual multi-species models to single-species models (Models 1–16; Supplementary Figs. 3–4). Estimates of arrowtooth flounder biomass and recruitment were larger in scale, but similar in trend, across models that included predation when compared to the single-species ensemble model. However, spawning stock biomass of arrowtooth flounder was lower across models that included predation. Estimates of biomass and spawning stock biomass of Pacific cod from the ensemble models were less sensitive to the inclusion of predation, although estimates of recruitment of Pacific cod were greater in ensemble models that included predation (Figs. 2–3).

3.5. Biomass consumed

Biomass consumed as prey of each species included in the model differed across model variants. When predation from Pacific halibut was included in the multi-species ensemble models as absolute numbers-at-age (Avg 3), biomass consumed as prey was relatively higher for all species (Fig. 4), leading to slightly lower estimates of biomass, spawning stock biomass, and recruitment. Trends were similar between multi-species ensemble models that excluded Pacific halibut predation (Avg 2) and treated Pacific halibut as relative numbers-at-age (Avg 4). The ensemble multi-species model that excluded predation from Pacific halibut (Avg 2) estimated that on average 661,560 mt of pollock, 29,072 mt of arrowtooth flounder, and 6249 mt of Pacific cod was consumed annually by predators included in the model (Fig. 4). However, the ensemble multi-species model that included predation from Pacific halibut as absolute numbers-at-age (Avg 3) estimated that on average 750,942 mt of pollock, 36,771 mt of arrowtooth flounder, and 7082 mt of Pacific cod was consumed annually by predators included in the model (Fig. 4). When Pacific halibut was included as relative numbers-at-age (Avg 4), average consumed biomass was estimated at 726,737 mt of pollock, 26,749 mt of arrowtooth flounder, and 6214 mt of Pacific cod, i.e. 110%, 92%, and 99% of that from Avg 2, respectively. Temporal trends in biomass consumed as prey were similar across ensemble models (Fig. 4).

Annual trends in biomass consumed as prey varied among species owing to trends in prey and predator abundance. Biomass of pollock consumed as prey peaked in 2014 coinciding with a peak in recruitment (Fig. 4). Similarly, peaks in biomass consumed as prey of arrowtooth flounder and Pacific cod coincided with elevated recruitment because younger individuals are more abundant and are more vulnerable to predation (Fig. 4). Depending on the model, consumption of arrowtooth flounder as prey peaked in 2000 or 2001 and of Pacific cod as prey peaked in 2006 or 2007 (Fig. 4). Across all model variants, arrowtooth flounder was the primary consumer of prey biomass in the model across all species, followed by Pacific halibut (Fig. 4). Predation mortality was the greatest for younger age-classes across all species, peaking between 2002 and 2006 owing to increased biomass of arrowtooth flounder.

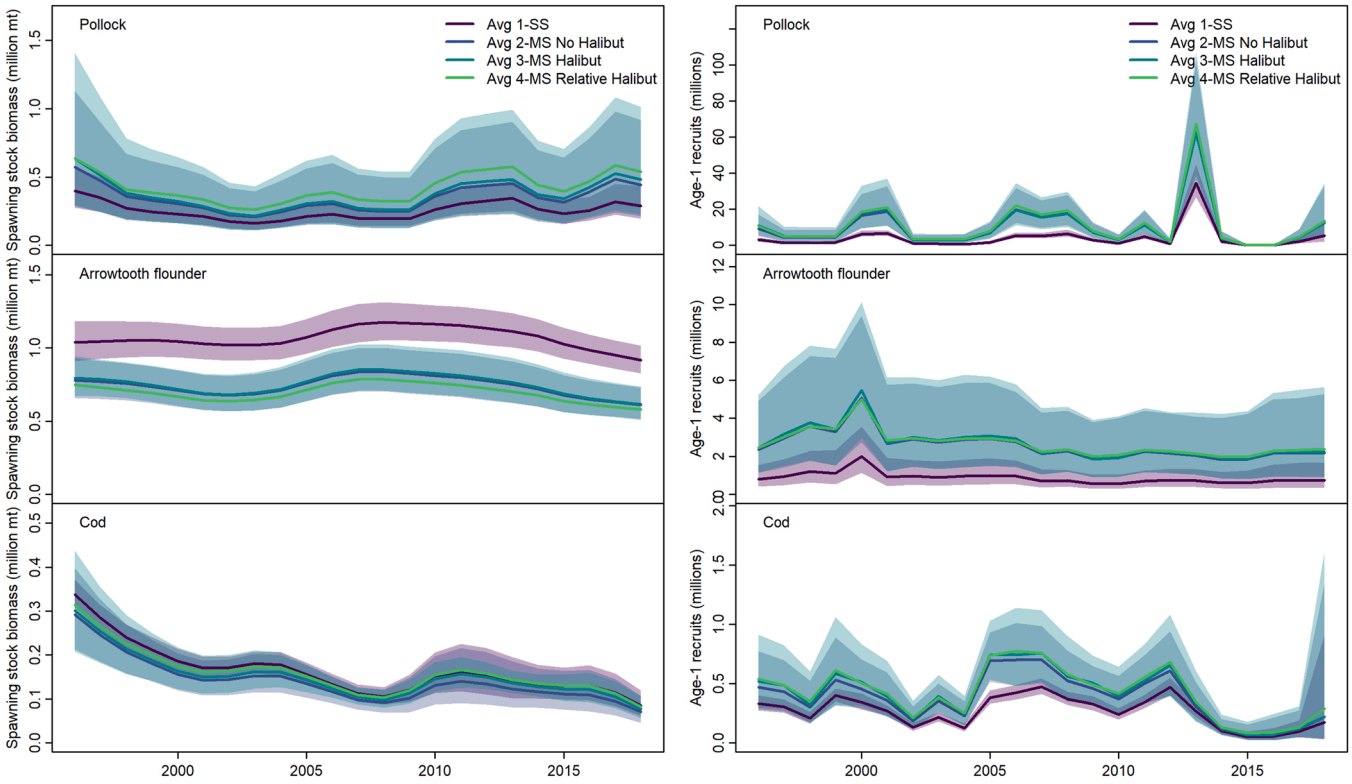


Fig. 3. Estimates of spawning stock biomass (left panel) and recruitment (right panel) and associated 95% confidence intervals (shaded region) from ensemble GOA CEATTLE models. See Table 5 for model descriptions. Note: estimates for Pacific halibut are excluded because they are not estimated/included in the model. Confidence intervals for Avg 4 are also excluded given high uncertainty.

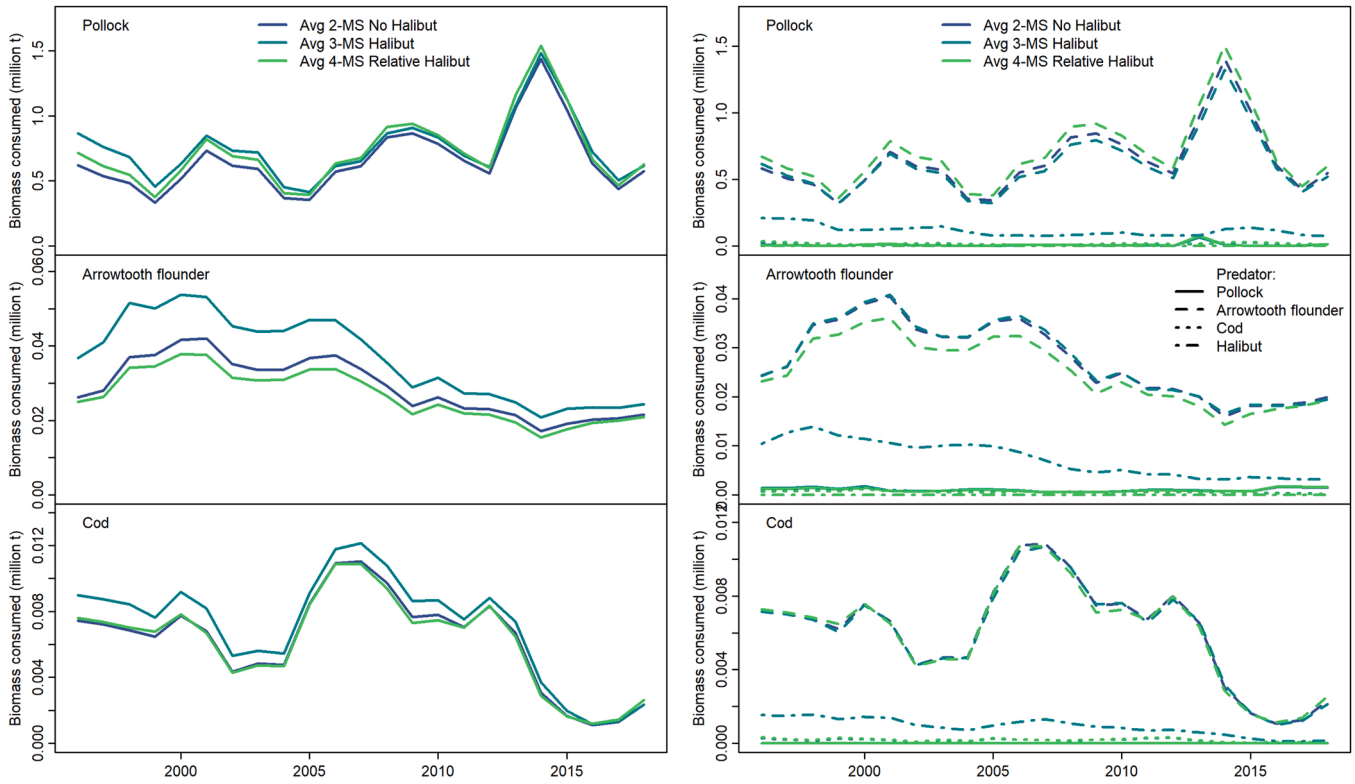


Fig. 4. Estimates of total biomass consumed as prey (left panel) and biomass consumed as prey by predator (right panel) from the ensemble GOA CEATTLE models. See Table 5 for model descriptions. For the right panel, solid lines indicate biomass consumed by pollock, dashed lines indicate biomass consumed by arrowthooth flounder, dotted lines indicate biomass consumed by cod, and dotted-dashed lines indicate biomass consumed by Pacific halibut.

Despite their smaller size, predation mortality by arrowtooth flounder males (and total natural mortality) was greater for males than for females. For example, on average across all years total natural mortality at age-1 was 1.4 yr^{-1} for pollock, 0.36 yr^{-1} for arrowtooth flounder females, 0.44 yr^{-1} for arrowtooth flounder males, and 0.95 yr^{-1} for Pacific cod in the ensemble model with absolute abundance-at-age of Pacific halibut (Avg 3; Fig. 4; Table S3). By age-5 total natural mortality of all species consisted mostly of M1, which was 0.45 yr^{-1} for pollock, 0.32 yr^{-1} for arrowtooth flounder females, 0.39 yr^{-1} for arrowtooth flounder males, and 0.47 yr^{-1} for Pacific cod as estimated from Avg 3 (Table S3).

3.6. Comparison of fixed vs relative halibut abundance

We developed models that used relative and absolute numbers-at-age for Pacific halibut to evaluate how pre-specifying predators in the model impacted the results. The scaling parameter when Pacific halibut was incorporated as relative numbers-at-age in IPHC Regulatory Areas 3A and 3B from the IPHC fishery-independent set-line survey was estimated with very high uncertainty to be 0.0021 (s.e. = 0.97) in Model 11 and 0.0025 (s.e. 1.44) in Model 16. The low estimate of this scaling parameter for Model 11 and 16 indicates that low Pacific halibut predation provided a better fit to the data. Biomass, spawning stock biomass, and recruitment of all species were similar between multi-species ensemble models that excluded Pacific halibut predation (Avg 2) and the multi-species model that included relative halibut numbers-at-age (Avg 4; Figs. 2–4). Similarly, time series trends of predation mortality for walleye pollock, Pacific cod, and arrowtooth flounder from Avg 2 and Avg 4 were very similar (Figs. 5–7). However, for cod and arrowtooth flounder, predation mortality was estimated to be greater and residual natural mortality (M1) was estimated to be lower in Avg 2 when compared to Avg 4 (Fig. 7).

4. Discussion

We found that perceptions of groundfish biomass in the GOA were sensitive to assumptions regarding predation. The GOA is a diverse and dynamic ecosystem that supports many commercial fisheries whose target species exhibit strong trophic interactions (Gaichas et al., 2015; Zador and Yasumiishi, 2016). Food web analysis has indicated that walleye pollock, Pacific cod, Pacific halibut, and arrowtooth flounder are highly ecologically connected and support some of the most valuable fisheries in the GOA (Barnes et al., 2020; Gaichas et al., 2015; Gaichas and Francis, 2008; Livingston et al., 2017; Thorson et al., 2019). While there is interest in developing models that can inform strategic and tactical EBFM of groundfish in the GOA, large scale spatial and temporal differences in the distribution of these species (primarily related to Pacific halibut) limits their utility to support the provision of management advice. We found that the MSCAA approach used here effectively integrates multiple forms of uncertainty regarding the distribution of Pacific halibut and the subsequent impacts of predation on groundfish in the GOA.

4.1. Extended MSCAA

The key forms of uncertainty we addressed related to the abundance and distribution of Pacific halibut in the GOA. However, rather than estimate the spatial distribution dynamics of Pacific halibut directly by fitting a habitat model to monitoring data, we input the absolute or relative numbers-at-age directly into the model. This builds upon previous approaches that have extended single- or multi-species models to account for predation from pre-specified total predator abundance (Hollowed et al., 2000; Spencer et al., 2016; van Kirk et al., 2012) by allowing for the impacts of predator age-structure on the dynamics of prey species. As recruited Pacific halibut are mainly considered predators in the system (Gaichas et al., 2010) and our study focused on

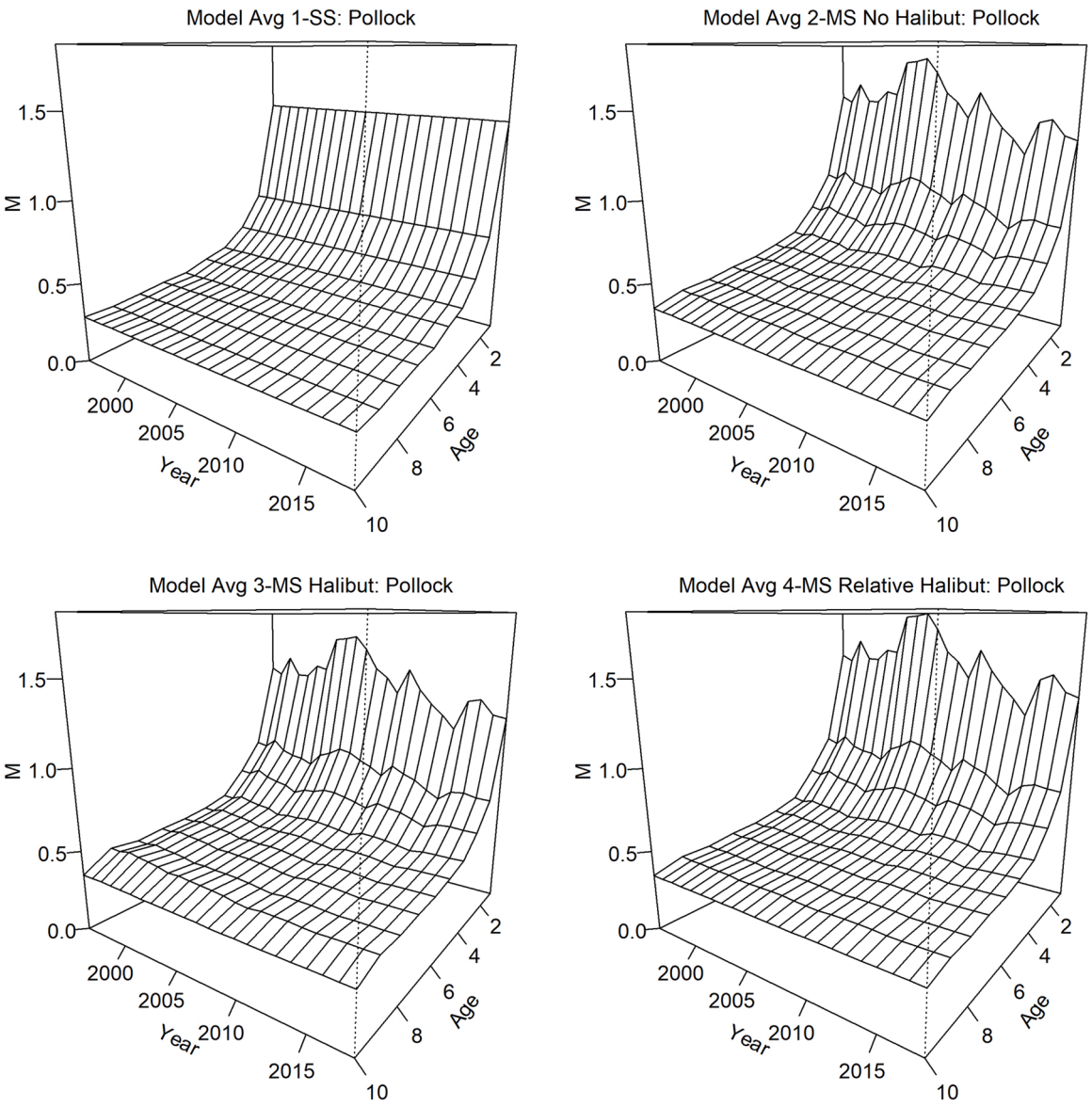


Fig. 5. Estimated total natural mortality (M1 + M2) for walleye pollock from ensemble models.

"top-down" predator dynamics, the extended multi-species approach allows us to simplify parameter estimation rather than recreate the dynamics of multiple complex stock assessment models used for management (Stewart and Hicks, 2019) while also addressing uncertainty in the predator numbers-at-age following the ensemble approach currently used for Pacific halibut management (Stewart and Hicks, 2019). Such an approach would not be appropriate for "bottom-up" predator-prey dynamics, such as prey-dependent reproduction, survival, and growth (Tulloch et al., 2018). However, this approach could be extended to other systems where "top-down" predation mortality is hypothesized to be a primary driver of target species' dynamics or where data limitations have historically precluded inclusion of predation in MICE models (e.g., marine mammals and fish). Additionally, the approach can be used to project the impact of future changes in "top-down" predation mortality given projected changes in predator-prey overlap or predator abundance.

4.2. Individual models

Currently, the Pacific halibut stock is assessed from California to Alaska using an ensemble of four equally weighted age-structured models (Stewart and Hicks, 2019) to account for uncertainty in the historical data available for Pacific halibut and the distribution of the fishery (Stewart and Martell, 2015). We developed multiple MSCAA models that included numbers-at-age from each IPHC model scaled by the proportion of the Pacific halibut stock in the GOA. Data on the proportion of the stock in the GOA are not available prior to 1993 and it is uncertain if recent decreases in the proportion of the Pacific halibut stock in the GOA reflect the historical distribution of Pacific halibut or the distribution in the absence of fishing (Stewart and Hicks, 2019). However, alternative assumptions regarding the historical distribution prior to 1993 relative to current trends of halibut in the GOA had little impact on the derived quantities. Similarly, derived quantities were relatively insensitive to which model was used to derive numbers-at-age of Pacific halibut. The insensitivity of the model results to parameterization of absolute numbers-at-age of Pacific halibut is likely due to the low relative contribution of halibut predation mortality compared to that of arrowtooth flounder. This is similar to results from Barnes et al. (2020), who found that arrowtooth flounder was the primary consumer of walleye pollock in the Gulf of Alaska. This is generally consistent with the relatively broad range of prey consumed by Pacific halibut (Barnes et al., 2018).

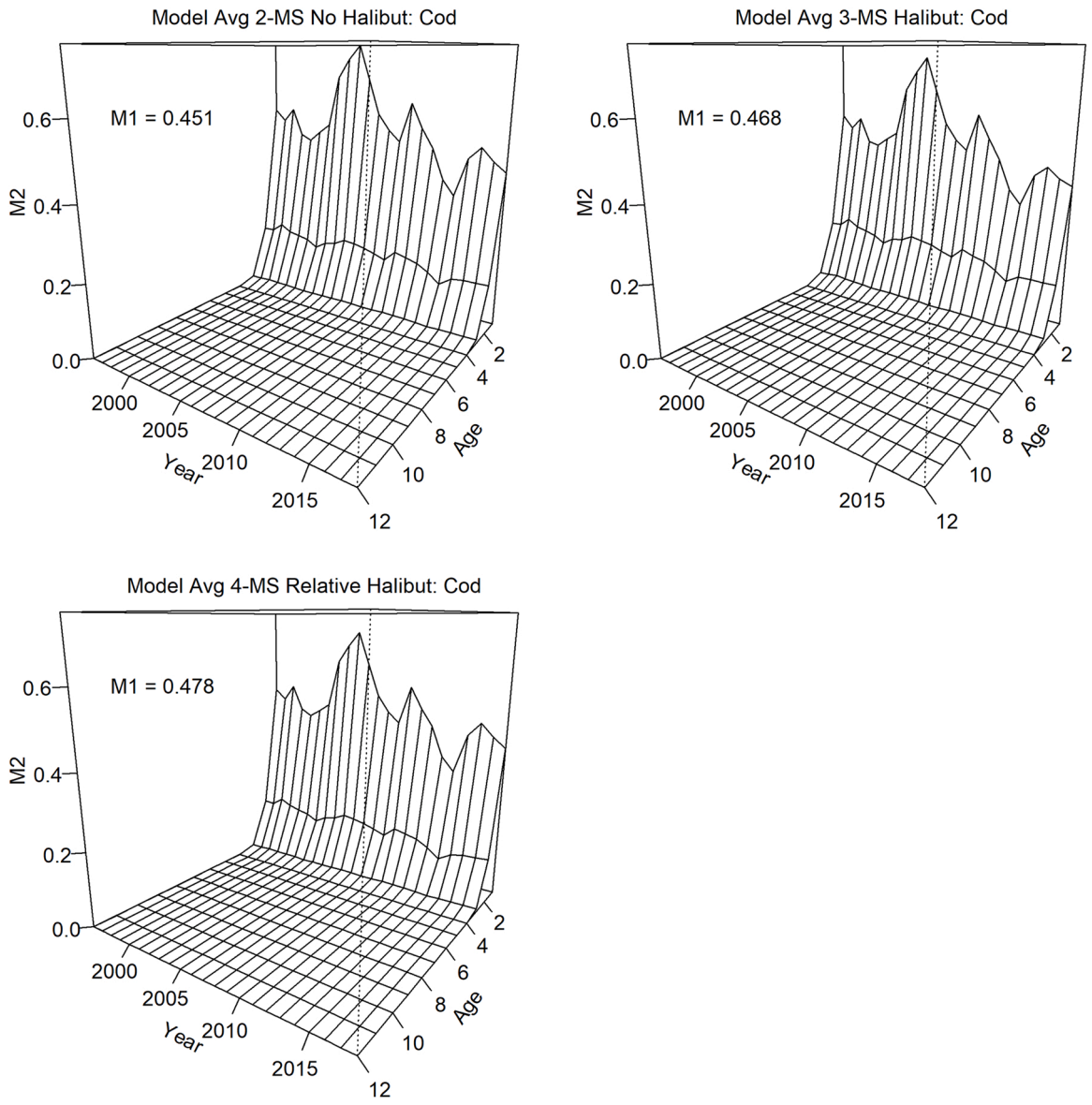


Fig. 6. Estimated residual mortality (M1) and predation mortality (M2) for Pacific cod from ensemble models that included predation. NOTE: For Model Avg 1 total natural mortality (M) is set to 0.504911 yr^{-1}.

Using an index of relative abundance-at-age derived from the IPHC setline survey in the GOA multiplied by an estimable scaling coefficient allowed us to evaluate alternative estimates of Pacific halibut in the GOA. This is similar to previous single-species approaches that estimated a scaling constant for predator abundance (Hollowed et al., 2000). Some caveats to this approach are that using the IPHC setline survey ignores the impacts of selectivity and scaling (i.e., catchability). The scaling coefficients were also estimated with a very high degree of uncertainty. Future research that aims to use indices of relative abundance-at-age would benefit by utilizing a prior distribution for the scaling coefficient or including additional data in the model likelihood to improve estimation. While this approach was used to evaluate alternatives to directly inputting estimates of absolute abundance at age, the small values of the estimated scaling coefficients for Models 11 and 16 led to results similar to models that excluded Pacific halibut predation. However, the increased flexibility in model parameterization allowed for slightly different fits to the data relative to Models 10 and 13, respectively. This approach may be useful for systems where data are not available to develop age-structured assessments for predators and rather indices of relative abundance are available, or cases where the spatial dynamics of a predator do not align well with those of the focal species.

Comparing models that included and excluded predation is a classic example of the bias-variance trade-off (Collie et al., 2016). Evaluating short-term models with fixed and relative numbers-at-age of halibut confirmed that the latter, more flexible approach provided a better fit to the data by eliminating Pacific halibut predation. The small value of the estimable scaling coefficient for Pacific halibut relative numbers-at-age effectively "shrunk" Pacific halibut predation. Similarly, multi-species models that excluded predation from Pacific halibut provided a better fit to the data and provided lower estimates of among and within model variance in derived quantities (as illustrated by the sensitivity of model outputs to time-series length and model averaged confidence intervals). Both examples are likely a consequence of overfitting the available data. Previous studies have demonstrated that models that do not include predation mortality may overfit data and perform poorly at predicting unobserved data (Trijoulet et al., 2019). Given that perceptions of recruitment and biomass are sensitive to model formation and the inclusion of predation, future research would benefit from evaluating the predictive performance of models developed here (Carvalho et al., 2021). This will also better affirm the need to account for the impacts of

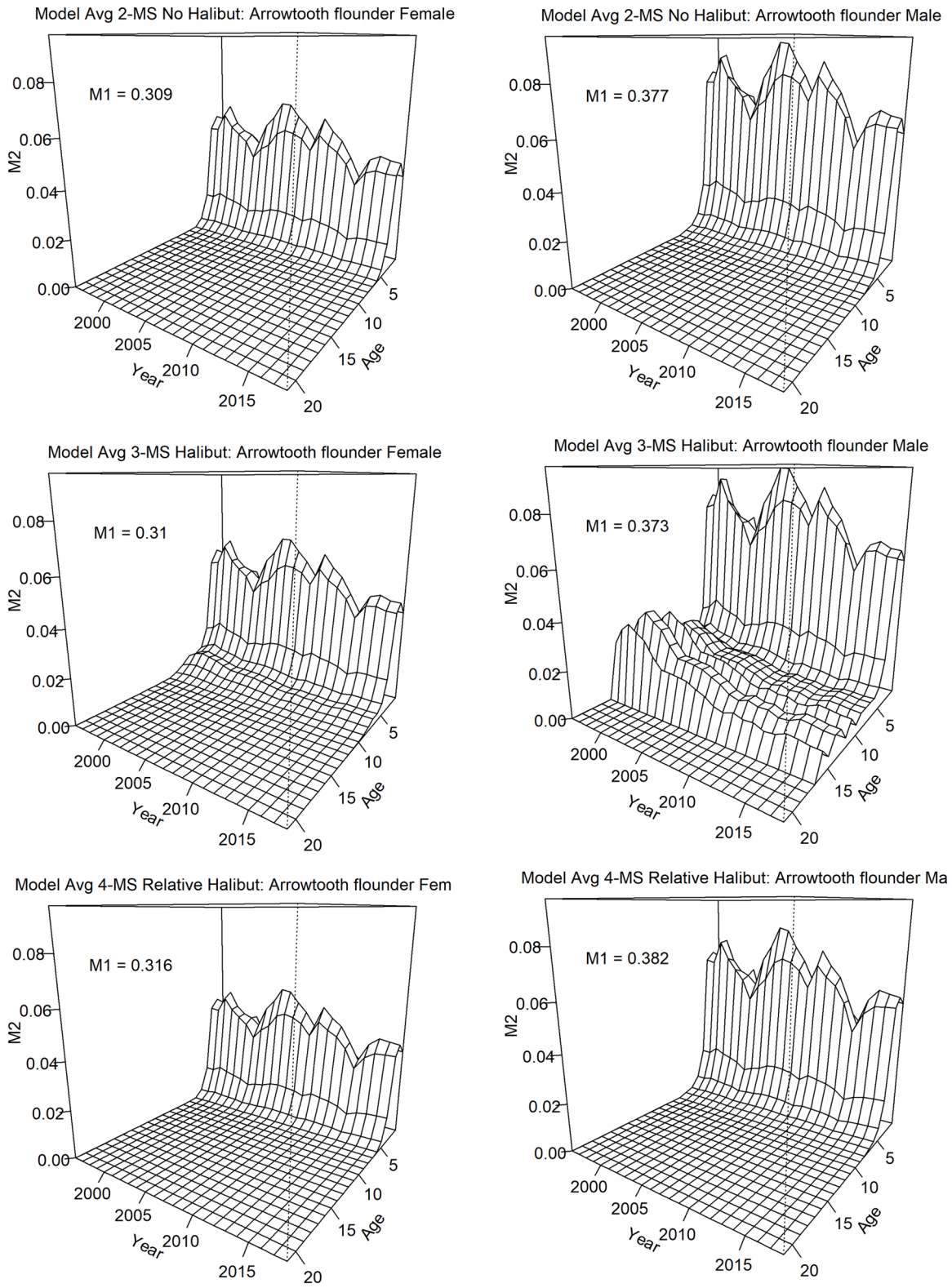


Fig. 7. Estimated residual mortality (M1) and predation mortality (M2) for Pacific cod from ensemble models that included predation. NOTE: For Model Avg 1 total natural mortality (M) is set to 0.2 yr^{-1} for females and 0.35 yr^{-1} for males.

time-varying predator-prey dynamics on target species' dynamics as clearly illustrated here, which suggests that increases in predator abundance leads to increased mortality of prey.

4.3. Ensembles

Rather than use a single "best" model for inference, an ensemble model approach was used to integrate across assumptions related to the abundance and distribution of Pacific halibut in the GOA. Ensemble

modelling approaches are gaining traction in fisheries assessment because of their ability to incorporate uncertainty in the state-of-nature and data available (Ianelli et al., 2016; Stewart and Martell, 2015; Zerbini et al., 2019). Derived quantities from ensemble models can be sensitive to the weighting scheme used and a variety of weighting schemes have been proposed: including information theoretic approaches (Symonds and Moussalli, 2011), Bayesian model averaging approaches (Zerbini et al., 2019), and approaches that rely on expert judgement (Stewart and Hicks, 2018). The data used for parameter estimation differed across individual models and, therefore, information theoretic and Bayesian model averaging approaches were not viable. To be consistent with the IPHC assessment we adopted a model averaging approach similar to theirs by giving equal weight to each model or input numbers-at-age (Stewart and Hicks, 2019). We did not average across predation dynamics because we were primarily interested in evaluating the impacts of predation and the inclusion of Pacific halibut in the assessment. However, future applications may benefit from integrating across states-of-nature in regard to predation.

4.4. Comparison to previous studies

Estimates of consumption from the GOA CEATTLE model differ slightly from those estimated from previous studies. Barnes et al. (2020) estimated that total consumption of walleye pollock by arrowtooth flounder, Pacific halibut, Pacific cod, and sablefish (*Anoplopoma fimbria*) to be 3.8 million metric tonnes in 1997 while Hollowed et al. (2000) (*sensu* Barnes et al., 2020) estimated that total consumption of walleye pollock by arrowtooth flounder, Pacific halibut, and stellar sea lions (*Eumetopias jubatus*) to be 3 million metric tonnes in 1997. We estimated consumption of walleye pollock to be 762,020 metric tonnes in 1997 by predators included in Avg 3. Apart from predators analyzed, differences in estimates of walleye pollock consumed may have resulted because Barnes et al. (2020) and Hollowed et al. (2000) used different methods to extrapolate predator and prey consumption rates to the population level, apotentially leading to larger estimates of predation than reported here in both cases. For example, Barnes et al. (2020) and Hollowed et al. (2000) did not include differences in prey age preferences across predator ages. Additionally, our model accounts for cannibalism despite lower estimates of consumed biomass of pollock, unlike Hollowed et al. (2000). As expected, given that these models were developed using similar datasets, we estimated that arrowtooth flounder was the primary consumer of the species included in the model, corroborating previous work in the GOA (Barnes et al., 2020; Gaichas et al., 2015; Hollowed et al., 2000).

Stock assessments for Pacific cod and arrowtooth flounder currently assume that natural mortality is age-invariant (Barbeaux et al., 2018; Spies and Palsson, 2018). However, when accounting for predation, total natural mortality is elevated for younger age classes, decreasing with age. In the case of Pacific cod, estimates of total natural mortality for older age classes of 0.451–0.478 yr^{-1} were slightly lower than the age-invariant M of 0.505 yr^{-1} (Barbeaux et al., 2018). MSCAA can be particularly useful by providing estimates of mean total natural mortality to be input into single-species stock assessments used for management (Dorn et al., 2018; ICES, 2019; Plagányi et al., 2022). Finally, because MSCAA models account for time- and age-varying natural mortality by explicitly modelling the time-varying dynamics of predator populations and age-specific predator suitability, outputs of time- and age-varying natural mortality can be used to identify how shifting predator-prey populations are impacting the dynamics of target species of interest.

4.5. Future

There are several assumptions utilized by the present modelling approach that may bias inference. First, the model assumes a Type II MSVPA functional response. Kinzey and Punt (2009) found that a predator pre-emption functional response best fit data from groundfish in the Aleutian Islands region. While comparing functional response models was beyond the scope of the present study, future research efforts would benefit by evaluating alternative functional response models given that model outputs are sensitive to the choice and parameterization of such models (Kinzey and Punt, 2009). Second, due to data limitations we used an estimate of average diet proportion to inform predation from multiple years and ignored variability (and uncertainty) in the diet data. Trijoulet et al. (2019) found that MSCAA were sensitive to the availability and quality of diet data. We used an average diet proportion estimate to account for biannual survey effort and data limitations prior to 1990. Comparing individual models that spanned different periods showed that they were relatively insensitive to the time-series of assessment and diet data used. However, future research using alternative methods to account for diet variability (e.g., using interpolation methods for years with missing data sensu Grüss et al., 2020) would be useful. Alternatively, diet data could be included in the model likelihood and suitability coefficients estimated.

Models developed here can be used to inform current and alternative tactical management procedures by showing responses arising from multi-species interactions. They can be used in management strategy evaluation approaches to account for the impact of multi-species interactions when evaluating the performance of alternative management strategies (Kaplan et al., 2021; Pérez-Rodríguez et al., 2022). I may also be plausible to design multi-species reference points from these models to better facilitate operationalizing EBFM (Holsman et al., 2020). Ecological reference points (e.g., Moffitt et al., 2016; Chagaris et al., 2020) and dynamic reference points (e.g., MacCall et al., 1985) can be derived from CEATTLE and provide alternative ways to guide fisheries management in a single-species context, accounting for predator-prey dynamics and variation in productivity, respectively. CEATTLE can also be used to inform reference point estimation under climate change (sensu Holsman et al., 2020). This is of particular importance given projected impacts of climate change and observed recent extreme events (e.g., the marine heatwave of 2014–2016; Barbeaux et al., 2020) and interactions with groundfish dynamics in the GOA (e.g., Cheung and Frölicher, 2020). Through the implementation of CEATTLE, trade-offs between harvesting of Pacific cod, pollock, arrowtooth flounder, and Pacific halibut under future climate conditions can be quantified. Such outputs, coupled with short- and long-term climate forecasts, can inform fisheries management advice that is robust to climate change and the apparent increases in environmental variability (Hollowed et al., 2020; Gaichas et al., 2017; Hazen et al., 2018; Kaplan et al., 2020; Holsman et al., 2019b; Karp et al., 2019).

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Credit authorship contribution statement

All authors have contributed significantly to the work, agreed to be listed, and approve the submitted version of the manuscript. Grant D. Adams developed the modelling and wrote the manuscript under the guidance and direction of Kirstin K. Holsman and André E. Punt and assistance of Steven J. Barbeaux, Martin W. Dorn, James N. Ianelli, Ingrid Spies, and Ian J. Stewart.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The Model runs, data and R code used in all population modelling can be found on GitHub (https://github.com/grantdadams/GOA-ceattle). An R package developed to implement the TMB CEATTLE model is available at GitHub (R Package: https://github.com/grantdadams/Rceattle).

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.fishres.2022.106303.

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